

Ring Formation

Planetary rings form predominantly from 3 different forces: gravitational, viscous and accretive interactions. For planetary rings the distance between significant particles is on the order of 200 m (NASA) and have masses on the order of 10^6 kg . Resulting forces are on the order of 10^{-3} N suggest this force is negligible, further that planetary rings are homogenous so any net gravitational interaction between ring particles is small.



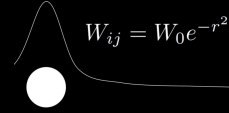
Finally, we model the viscous forces, which turn out to be the predominant force in ring formation (qualitative observation from simulation). For a fluid the viscous force is $F = \mu \nabla^2 v$ (Wikimedia, 2024) which for a smooth particle is $F = \mu \sum_j m_j \frac{v_i - v_j}{\rho_j} W_{ij}$ (Steve Rotenberg, 2019.) Where W_{ij} is the smoothing kernel. In our case we want $W_{ij} \propto \rho_j$ so we must find $\rho_j(r, \theta, \phi) = \rho_j(r)$ for our particle. The hydrostatic equation (Atlantic Oceanographic and Meteorological Agency.) is $\frac{dp}{dr} = -\rho(r)g(r)$ where $g(r)$ is the acceleration, in this case due to gravity. Therefore, $\frac{dp}{dr} = -\frac{G\rho(r)}{r^2} \int_V \rho(r) d\Omega = -\frac{G\rho(r)}{r^2} \cdot 4\pi \int r^2 \rho(r) dr$.

Which is rearranged to give $\frac{dp}{dr} \cdot \frac{r^2}{\rho(r)} = -4\pi G \int r^2 \rho(r) dr \implies \frac{d}{dr} \left(\frac{r^2 p}{\rho(r)} \right) = -4\pi G r^2 \rho(r)$. Using ideal gas law is $p = \rho(r) \frac{k_B T}{\mu m}$ and substituting in $\frac{d}{dr} \left(\frac{r^2 p}{\rho(r)} \right) = -\frac{\mu m}{k_B T} 4\pi G r^2 \rho(r)$.

Accretion can be modeled by considering when two bodies have overlapping Hill spheres (Crida & Charnoz



2012). Such bodies then have some probability of accretion depending on the ratio $\frac{\Delta v}{v}$ of the bodies relative velocities (Ohtsuki 1993).



To solve this differential equation use the ansatz $\rho = \rho_0 e^{\sum_k \alpha_k x^k}$. Then $\frac{d\rho(r)}{dr} = \rho_0 \sum_k \alpha_k k r^{k-1} \cdot e^{\sum_k \alpha_k x^k} = \rho(r) \sum_k \alpha_k k r^{k-1}$. Then the LHS is $\frac{d}{dr} \left(\frac{r^2 p}{\rho(r)} \right) = \frac{d}{dr} \left(\rho(r) \sum_k \alpha_k k r^{k-1} \frac{r^2}{\rho(r)} \right) = \frac{d}{dr} \left(\sum_k \alpha_k k r^{k+1} \right) = \sum_k \alpha_k k(k+1) r^k$. And, the RHS is $-\frac{\mu m}{k_B T} 4\pi G r^2 \rho(r)$. Considering $\alpha = \alpha_2$ then in the small r limit RHS is $-\frac{\mu m}{k_B T} 4\pi G r^2 \rho(r) \approx -\frac{\mu m}{k_B T} 4\pi G r^2 \rho_0 (1 + \alpha r^2) = -\frac{\mu m}{k_B T} 4\pi G \rho_0 (r^2 + \alpha r^4) \approx -\frac{\mu m}{k_B T} 4\pi G \rho_0 r^2 \approx 6\alpha r^2$. Therefore, $\alpha = -\frac{1}{6} \frac{\mu m}{k_B T} 4\pi G \rho_0$.

Applying this to our equation for the viscous force using $W_{ij} \propto \rho \implies W_{ij} = W_0 e^{-\alpha r^2}$. So the interaction force is $F_{ij} = \frac{p_i - p_j}{\rho_0} W_0 e^{-\alpha r^2}$.